

Legendre's Equation and Legendre's Functions

Question: Explain Legendre's equation and Legendre's functions of the first kind $P_n(x)$ and second kind $Q_n(x)$.

Legendre's Equation

The differential equation

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n + 1)y = 0$$

where n is a non-negative integer, is called **Legendre's differential equation**. It is a second-order linear differential equation and has two independent solutions, $P_n(x)$ and $Q_n(x)$.

Legendre's Function of the First Kind $P_n(x)$

One solution of Legendre's equation can be expressed as

$$y = a_0 \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \dots \right].$$

where a_0 is an arbitrary constant.

If n is a positive integer and

$$a_0 = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!},$$

the above solution becomes

$$P_n(x) = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \dots \right].$$

The solution is called the **Legendre polynomial** or **Legendre function of the first kind**, denoted by $P_n(x)$.

Legendre's Function of the Second Kind $Q_n(x)$

The second independent solution of Legendre's equation is called the **Legendre's function of the second kind**, denoted by $Q_n(x)$.

In the descending-power form:

$$Q_n(x) = \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n+1)} \left[x^{-n-1} + \frac{(n+1)(n+2)}{2(2n+3)} x^{-n-3} + \dots \right].$$

Unlike $P_n(x)$, the series for $Q_n(x)$ is **non-terminating**.